

CHAPTER 5

5.1) CENTRIFUGE

The centrifugal force on a gas particle is Mw^2r . The corresponding potential energy relative to the axis of rotation is $-Mw^2r^2/2$, which serves as external chemical potential. Hence the total chemical potential is

$$\mu = \tau \log[n(r)/n_Q] - Mw^2r^2/2 \quad ,$$

which is constant (independent of r) if

$$n(r) = n(0) \exp(Mw^2r^2/2\tau) \quad .$$

Note that wr is the peripheral speed of the centrifuge: The exponential factor is the inverse Boltzmann factor for the kinetic energy of a particle with the velocity v , $\exp(Mv^2/2\tau)$.

Comments. Centrifuging is an effective way to separate isotopes, or to precipitate heavy molecules from solutions. For a gas containing two isotopes with a mass difference ΔM , the heavy-to-light isotope ratio is increased along the perimeter by a factor $\exp(\Delta Mv^2/2\tau)$ relative to the center, or conversely, the light-to-heavy isotope ratio is increased by this factor at the center. In such gas centrifuges, peripheral speeds of 350 m s^{-1} have been achieved. For the two uranium isotopes ^{238}U and ^{235}U , with $\Delta M = 3 \text{ amu}$, this speed leads to a theoretical enrichment factor of 1.076, or 7.6 percent. In this case the gas used is UF_6 , and the desired rare isotope ^{235}U accumulates near the axis. For several practical reasons the enrichment factor actually achieved is substantially less than this theoretical ideal.

In the precipitation of heavy organic molecules the mass difference ΔM is the mass different between the molecule

and the solvent it displaces. This mass difference can be manipulated over some range, by changing the density of the solvent, by adding a salt, or by similar means.

5.2) MOLECULES IN THE EARTH'S ATMOSPHERE

The potential energy of a molecule in the Earth's gravitational field is $-MgR^2/r$, taken at distance r from the center of the earth, and relative to $r = \infty$. The total chemical potential is therefore

$$\mu(r) = \tau \log[n(r)/n_Q] - MgR^2/r \quad .$$

If this is to be independent of r , $\mu(r) = \mu(R)$, we must have

$$\tau \log[n(r)/n(R)] = MgR^2/r - MgR$$

or

$$n(r) = n(R) \exp(-MgR/\tau) \exp(MgR^2/r\tau) \quad .$$

The total number of molecules in the atmosphere is

$$\begin{aligned} N &= 4\pi \int_R^\infty n(r) r^2 dr \\ &= 4\pi n(R) \exp(-MgR/\tau) \int_R^\infty \exp(MgR^2/r\tau) r^2 dr \quad . \end{aligned} \quad (75)$$

For $r \rightarrow \infty$, the exponential approaches 1, and the integrand becomes r^2 , which diverges, so that an equilibrium state with finite N does not exist. Molecules with a kinetic energy $\frac{1}{2}Mv^2 > MgR$ may escape if they do not collide with other molecules. For the lightest of all molecules, H_2 , one finds $MgR = 2 \times 1.67 \times 10^{-24} g \times 981 \text{ cm s}^{-2} \times 6.37 \times 10^8 \text{ cm} \cong 2 \times 10^{-12} \text{ erg}$. At 300 K, the mean kinetic energy of the molecules is $(3/2)k_B \times 300 \text{ K} \cong 6.2 \times 10^{-14} \text{ erg}$. Hence, only a very small fraction of the molecules has the high kinetic energy necessary to escape; even the lightest molecules

will escape only very slowly.

5.3) POTENTIAL ENERGY OF GAS IN GRAVITATIONAL FIELD

From (18), $n(h) = n(0) \exp(-Mgh/\tau)$, one obtains the total number of atoms in a column, per unit cross-sectional area,

$$N = \int_0^{\infty} n(h) dh = n(0)\tau/Mg \quad .$$

The total potential energy of the atoms in the column is

$$\begin{aligned} U_{\text{pot}} &= \int_0^{\infty} n(h) Mgh \, dh \\ &= n(0)\tau^2/(Mg) \int_0^{\infty} e^{-x} x \, dx = n(0)\tau^2/Mg = N\tau \quad . \end{aligned}$$

The average potential energy per atom is τ . The average kinetic energy is $3\tau/2$, hence the average total energy is $5\tau/2$ and the heat capacity $C = 5/2$.

Comment. The heat capacity is the familiar value for C_p , the heat capacity at constant pressure. Of course: As the gas is heated it expands upward in the gravitational column. This requires work just like work against a piston that keeps the pressure constant. The pressure at $h = 0$ is indeed kept constant: It is simply the constant weight per unit area of the gas column above.

5.4) ACTIVE TRANSPORT

$$\begin{aligned} \Delta\mu &= \mu(\text{sap}) - \mu(\text{pond}) = \tau \log[n(\text{sap})] - \tau \log[n(\text{pond})] \\ &= \tau \log[n(\text{sap})/n(\text{pond})] = \tau \log 10^4 \quad . \end{aligned}$$

At $T = 300\text{K}$: $\tau = 0.0259 \text{ eV}$. Hence $\tau = 0.238 \text{ eV}$. Because potassium is singly ionized, this corresponds to a voltage

of 0.238 V.

5.5) MAGNETIC CONCENTRATION

The lines for $B = 0$ and $B = 20 \text{ kG} = 2\text{T}$ are shifted horizontally by a concentration ratio $n(B)/n(0) = 100$. From (25), for $n(B) \gg n(0)$,

$$\begin{aligned} m/\tau &= \cosh^{-1}[n(B)/n(0)]/B \cong \log[2n(B)/n(0)]/B \\ &= \log 200/2 \times 10^4 \text{G} = 2.65 \times 10^{-4} \text{G}^{-1} = 2.65 \text{ T}^{-1} , \\ m/\mu B &= (m/\tau)(k_B T/\mu_B) \\ &= 2.65 \times 10^{-4} \text{G}^{-1} \times 1.38 \times 10^{-16} \text{erg K}^{-1} / (9.27 \times 10^{-21} \text{erg K}^{-1}) \\ &\cong 1183 \text{ (= average number of elementary magnets)} \end{aligned}$$

5.6) GIBBS SUM FOR A TWO LEVEL SYSTEM

(a) There are three states:

Occupancy N:	0	1	1
Energy:	0	0	ε

$$\mathcal{Z} = 1 + \lambda + \lambda \exp(-\varepsilon/\tau) . \quad (76)$$

$$\begin{aligned} \text{(b) } \langle N \rangle &= \frac{1}{\mathcal{Z}} [0 \times 1 + 1 \times \lambda + 1 \times \lambda \exp(-\varepsilon/\tau)] \\ &= (\lambda/\mathcal{Z}) [1 + \exp(-\varepsilon/\tau)] \end{aligned} \quad (77)$$

(c) The thermal average occupancy of the state at energy ε is given by the last term in the expression for $\langle N \rangle$,

$$\begin{aligned} \langle N(\varepsilon) \rangle &\cong \frac{\lambda}{\mathcal{Z}} \exp(-\varepsilon/\tau) . \\ \text{(d) } U &= \frac{1}{\mathcal{Z}} [0 \times 1 + 0 \times \lambda + \varepsilon \times \lambda \exp(-\varepsilon/\tau)] \\ &= (\varepsilon \lambda / \mathcal{Z}) \exp(-\varepsilon/\tau) = \langle N(\varepsilon) \rangle \varepsilon . \end{aligned} \quad (78)$$

(e) Adding a state with occupancy $N = 2$ and energy $\varepsilon + 0 = \varepsilon$ contributes an additional term $\lambda^2 \exp(-\varepsilon/\tau)$ to $\mathcal{Z}$:

$$\begin{aligned}\mathcal{Z} &= 1 + \lambda + \lambda \exp(-\varepsilon/\tau) + \lambda^2 \exp(-\varepsilon/\tau) \\ &= (1+\lambda)[1+\lambda \exp(-\varepsilon/\tau)].\end{aligned}\quad (79)$$

This has the form $\mathcal{Z} = \mathcal{Z}_0 \mathcal{Z}_\varepsilon$ of the Gibbs sum for two independent systems, one with the fixed energy 0 independent of occupancy, and one with the energy 0 if empty and ε if occupied. Both systems can be occupied with either zero or one particle, and the occupancy of one is independent of the occupancy of the other.

5.7) STATES OF POSITIVE AND NEGATIVE IONIZATION

N :	0	1	1	2
ε :	$-\delta/2$	$-\Delta/2$	$+\Delta/2$	$+\delta/2$

$$\mathcal{Z} = \exp(\delta/2\tau) + \lambda[\exp(\Delta/2\tau) + \exp(-\Delta/2\tau)] + \lambda^2 \exp(-\delta/2\tau)$$

$$\langle N \rangle = 0 + \lambda[\exp(\Delta/2\tau) + \exp(-\Delta/2\tau)] + 2\lambda^2 \exp(-\delta/2\tau).$$

To obtain $\langle N \rangle = 1$ the right-hand sides of the two last lines must be equal: $\lambda^2 \exp(-\delta/2\tau) = \exp(+\delta/2\tau)$, or

$$\lambda^2 = \exp(\delta/\tau) \quad .$$

Because of (60) this implies $\mu = \delta/2$. The quantity Δ has dropped out.

5.8) CARBON MONOXIDE POISONING

With the abbreviations $x = \lambda_A \exp(-\varepsilon_A/\tau)$ and $y = \lambda_B \exp(-\varepsilon_B/\tau)$ we have the Gibbs sum $\mathcal{Z} = 1 + x + y$. The fraction of sites occupied by O_2 is

$$f = x/\mathcal{Z} = x/(1+x+y) \quad . \quad (S1)$$

(a) No CO means $\lambda_B = 0$, $y = 0$; hence $x = f/(1-f) = 0.9/0.1 = 9$, and $\varepsilon_A = -\tau \log(x/v_A) = -\tau \log(9 \times 10^5) = -13.7 \tau = 0.366 \text{ eV}$, where we have used $\tau = 8.62 \times 10^{-5} \text{ eV K}^{-1} \times 310 \text{ K} = 0.0267 \text{ eV}$.

(b) We solve (S1) for y : $y = x/f - (1+x) = 80$. Thus, $\varepsilon_B = -\tau \log(y/\lambda_B) = -\tau \log(80 \times 10^7) = -20.5\tau = -0.548 \text{ eV}$. We see that CO is more tightly bound by about 6.8τ . This means that a relatively small concentration of CO can make the binding sites inaccessible to O_2 . This is why CO is such a poison.

5.9) ADSORPTION OF O_2 IN A MAGNETIC FIELD

The magnetic moment may be parallel, perpendicular, or antiparallel to the magnetic field. Hence,

$$\mathcal{Z} = 1 + \lambda \{ \exp[-(\varepsilon - \mu_B)/\tau] + \exp(-\varepsilon/\tau) + \exp[-(\varepsilon + \mu_B)/\tau] \}$$

$$\mathcal{Z} = 1 + \lambda \exp(-\varepsilon/\tau) [1 + 2 \cosh(\mu_B/\tau)] ,$$

where ε is the energy of an occupied site when $B = 0$. If we define $x = 3\lambda \exp(-\varepsilon/\tau)$, $1+\delta = [1 + 2 \cosh(\mu_B/\tau)]/3$, we may write the occupancy as

$$f = \frac{x(1+\delta)}{1+x(1+\delta)} = \frac{x}{1+x} \frac{1+\delta}{1+\frac{x}{1+x}\delta} = f_0 \frac{1+\delta}{1+f_0\delta} ,$$

where $f_0 = x/(1+x)$ is the occupancy when $B = 0$. For small changes δ , we may approximate

$$(1+\delta)/(1+f_0\delta) \cong 1 + (1-f_0)\delta .$$

To change the occupancy f by one percent (relative to $f_0 = 0.9$), we must have $(1-f_0)\delta = 0.01$, or $\delta = 0.1$. For small δ and hence small μ_B/τ we may expand the hyperbolic cosine, $\cosh(\mu_B/\tau) \cong 1 + \frac{1}{2}(\mu_B/\tau)^2$, hence

$$\delta \cong \frac{2}{3} [\cosh(\mu_B/\tau) - 1] \cong \frac{1}{3} (\mu_B/\tau)^2 \cong 0.1 ,$$

$$\mu_B B / \tau \cong 0.3^{\frac{1}{2}} \cong 0.548 \quad ,$$

$$B = 0.548 \tau / \mu_B \cong 2.4 \times 10^6 \text{ Gauss} \quad .$$

This number should dispense with any notions that technologically achievable magnetic fields have detectable effects on the respiratory process.

5.10) CONCENTRATION FLUCTUATIONS

(a) From (52) and (53), omitting the subscript $s(N)$ on ε ,

$$\begin{aligned} \langle N^2 \rangle &= \sum_{ASN} N^2 P(N) = \frac{1}{\mathcal{Z}} \sum_{ASN} N^2 \exp[(N\mu - \varepsilon)/\tau] \\ &= \frac{\tau^2}{\mathcal{Z}} \sum \frac{\partial^2}{\partial \mu^2} \exp[(N\mu - \varepsilon)/\tau] = \frac{\tau^2}{\mathcal{Z}} \left(\frac{\partial^2 \mathcal{Z}}{\partial \mu^2} \right)_{\tau} \quad . \end{aligned}$$

(b) From (80),

$$\left(\frac{\partial \langle N \rangle}{\partial \mu} \right)_{\tau} = \frac{\tau}{\mathcal{Z}} \left(\frac{\partial^2 \mathcal{Z}}{\partial \mu^2} \right) - \frac{\tau}{\mathcal{Z}^2} \left(\frac{\partial \mathcal{Z}}{\partial \mu} \right)_{\tau}^2 = \frac{1}{\tau} \langle (\Delta N)^2 \rangle \quad ,$$

where the last equality follows by comparison with (82).

5.11) EQUIVALENT DEFINITION OF CHEMICAL POTENTIAL

From (31), for a change at constant volume ($dV = 0$),

$$d\sigma = (\partial \sigma / \partial U)_{N,V} dU + (\partial \sigma / \partial N)_{U,V} dN = \frac{1}{\tau} dU - \frac{\mu}{\tau} dN \quad ,$$

where we have used (35). Solved for dU ,

$$dU = \tau d\sigma + \mu dN \quad .$$

But if $U = U(\sigma, V, N)$, we must also have, when $dV = 0$,

$$dU = (\partial U / \partial \sigma)_{V,N} d\sigma = (\partial U / \partial N)_{\sigma,V} dN \quad .$$

Comparing the two expressions yields

$$\mu = (\partial U / \partial N)_{\sigma,V} \quad .$$

5.12) ASCENT OF SAP IN TREES

The chemical potential of water molecules in air with a relative humidity $r < 1$ differs from that in air saturated with water vapor ($r = 1$) by

$$\Delta\mu_{\text{int}} = \tau [\log(rn/n_Q) - \log(n/n_Q)] = \tau \log r < 0 \quad .$$

This is an internal chemical potential difference driving the evaporation process through the leaf surface. In steady state, any water molecule evaporating at height h must be replaced by a molecule coming up from the roots, acquiring a potential energy $\Delta\mu_{\text{ext}} = Mgh$ in the process. It acts as an external chemical potential step. The evaporation will cease when the diffusive equilibrium condition (16) has been reached:

$$\Delta\mu_{\text{ext}} = - \Delta\mu_{\text{int}} \quad ,$$

$$h = \frac{\tau}{Mg} \log (1/r) = \frac{k_B \times 298 \text{ K} \times 0.105}{18 \text{ amu} \times 981 \text{ cm s}^{-2}} \cong 1470 \text{ meter} \quad .$$

Comments. The result would be modified by numerous complications that we have omitted for simplicity. For example: Most trees do not stand in water but in more or less moist soil; there is usually a temperature difference between the roots and the leaves; the water inside the tree contains dissolved substances that change its chemical potential, etc.

The problem offers an opportunity for a classroom discussion of the following problem. The water column inside the tree is clearly in equilibrium with itself. But because the external chemical potential increases with height, the internal chemical potential must decrease. This decrease is the decrease caused by the decrease of hydrostatic pressure with height. The chemical potential depends on pressure according to the Maxwell relation

$$\left(\frac{\partial \mu}{\partial p}\right)_T = \left(\frac{\partial \mu}{\partial N}\right)_T = \frac{1}{n} ,$$

obtained by cross-differentiation of the thermodynamic identity in the form

$$dG \equiv d(F+pV) = - \sigma d\tau + Vdp + \mu dN .$$

Above a height of about 9.81 meters, the water is actually under negative pressure. Water confined to microscopic capillaries as in a tree can withstand large negative pressures before bubbles can form. The process of nucleating bubbles in a liquid is similar to the process of nucleating droplets in a vapor, discussed in Chapter 10.

For two reviews on negative pressure in liquids see (a) A.J.T. Hayward, Am. Scientist 59(#4), 434-443, (Jul./Aug. 1971); (b) R.E. Apfel, Scientific American 227(#12), 58-71 (Dec. 1972).

5.13) ISENTROPIC EXPANSION

(a) From (3.68), $Z_N = (Z_1)^N / N!$, where N is the number of particles and

$$Z_1 = \sum \exp[-\varepsilon(\vec{n})/\tau]$$

is the single particle partition function. The vector $\vec{n}$ designates the orbital. The occupancy of orbital $\vec{n}$ is

$$f(\vec{n}) = (N/Z_1) \exp[-\varepsilon(\vec{n})/\tau] .$$

Next, from (3.49) and (3.55):

$$\begin{aligned} \sigma &= - \frac{\partial F}{\partial \tau} = + \frac{\partial}{\partial \tau} (\tau \log Z) \\ &= N \log Z - \log N! + N\tau \frac{\partial}{\partial \tau} \log Z . \end{aligned}$$

The last term can be written

$$\begin{aligned}
\frac{N\tau}{Z_1} \sum \frac{\varepsilon(\vec{n})}{\tau^2} \exp[-\varepsilon(\vec{n})/\tau] &= \sum \frac{\varepsilon(\vec{n})}{\tau} f(\vec{n}) \\
&= \sum [\log N - \log Z_1 - \log f(\varepsilon_n)] f(\vec{n}) \\
&= N(\log N - \log Z_1) - \sum f(\vec{n}) \log f(\vec{n}).
\end{aligned}$$

With this

$$\sigma = N \log N - \log N! - \sum f(\vec{n}) \log f(\vec{n}).$$

But, from Stirling's formula, (A.29), $N \log N - \log N!$
 $= N = \sum f(\vec{n})$, hence

$$\sigma = \sum f(\vec{n}) [1 - \log f(\vec{n})]. \quad (S1)$$

(b) From (3.59), all orbital energies depend on the volume like $\varepsilon \propto L^{-2} = V^{-2/3}$. Hence, the partition function and with it the occupancies depend on τ and V only through the product $\tau V^{2/3}$. If this product is constant, the occupancies will be constant and therefore, because of (S1), the entropy will be constant.

Comment. The result (b) could of course, also be obtained directly from the Sackur-Tetrode equation, (3.76). Because of (3.63): $n_Q/n = n_Q V/N \propto V\tau^{3/2}$. Thus, $\sigma = \text{constant}$ requires $V\tau^{3/2} = \text{constant}$, which is equivalent to $\tau V^{2/3} = \text{constant}$. We prefer the occupancy argument because it is more "physical": The entropy remains constant because the occupancies remain constant.

Students insisting on mathematical rigor might object to our treatment under (b) that we have only shown that $\tau V^{2/3} = \text{constant}$ leads to $\sigma = \text{constant}$, but not necessarily the inverse. The Sackur-Tetrode argument is immune to this criticism.

5.14) MULTIPLE BINDING OF O₂

We assume that the four O₂ binding sites are independent. The Gibbs sum for a single binding site would be

$$\mathcal{Z}_1 = 1 + \lambda \exp(-\varepsilon/\tau).$$

For four independent binding sites

$$\begin{aligned} \mathcal{Z}_4 &= (\mathcal{Z}_1)^4 = [1 + \lambda \exp(-\varepsilon/\tau)]^4 \\ &= 1 + 4\lambda \exp(-\varepsilon/\tau) + \dots + \lambda^4 \exp(-4\varepsilon/\tau). \end{aligned}$$

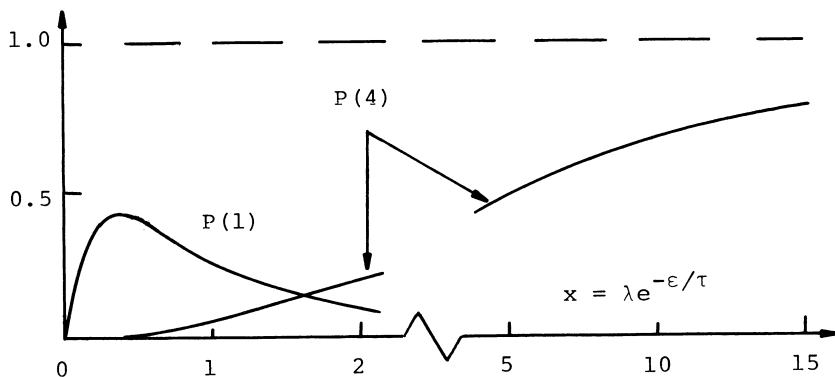
Here the first term corresponds to no occupancy, the second term to occupancy of one of the binding sites, and the last term to occupancy of all four binding sites.

$$(a) \quad P(1) = 4\lambda \exp(-\varepsilon/\tau) / \mathcal{Z} = 4x/(1+x)^4,$$

where $x = \lambda \exp(-\varepsilon/\tau)$. The probability starts linearly with λ , goes through a maximum $P(1) = 27/64$ at $x = 1/3$, and then falls off towards zero.

$$(b) \quad P(4) = \lambda^4 \exp(-4\varepsilon/\tau) / \mathcal{Z} = [x/(1+x)]^4.$$

This starts proportionally to λ^4 , then levels out slowly and approaches $P(4) = 1$ asymptotically.



5.15) EXTERNAL CHEMICAL POTENTIAL

(a) Moving the system to a height h without a change in the state of the system otherwise, adds the potential energy $NMgh$ to the state of the system. This added energy is the same for all states of the system, hence we add a factor $\exp(-NMgh/\tau)$ to every term in the partition function, and therefore to the partition function itself,

$$Z(h) = Z(0) \times \exp(-NMgh/\tau).$$

From (3.55) we obtain the Helmholtz free energy:

$$F(h) = -\tau \log Z(h) = -\tau \log Z(0) + NMgh = F(0) + NMgh.$$

From this and the definition (5) of the chemical potential, by differentiation with respect to N ,

$$\mu(h) = \mu(0) + Mgh.$$

(b) In the barometric problem $\mu(h) = \mu(0)$ because the two systems were in diffusive equilibrium with each other. If we simply lift a system by h it would not be in diffusive equilibrium with an otherwise identical system left behind on the ground: On diffusive contact, particles would flow from the upper to the lower system until the chemical potentials have equilibrated.

SUPPLEMENTARY MATERIAL TO CHAPTER 5

VAPOR PRESSURE AS MEASURE OF CHEMICAL POTENTIAL

Consider any liquid or solid substance that contains the atomic or molecular species A. Suppose that $p_A(\tau)$ is the equilibrium vapor pressure of A of the substance. That is, if the substance is placed into an initially evacuated enclosed volume, a certain amount of one or more of the constituents of the substance will evaporate until a certain equilibrium pressure has built up -- which may or may not have a measurable magnitude. The pressure p_A is the partial pressure of the constituent species A.

Equilibrium means that the chemical potential of μ_A of species A in the substance must equal the chemical potential μ_A of A in the vapor. If the equilibrium vapor pressure is sufficiently small, the vapor may be treated as an ideal gas, and (12b) must be valid, in the form

$$\mu_A = \tau \log(p_A/\tau n_{QA}) \quad .$$

This means that equilibrium vapor pressure measurements can be used to measure chemical potential. For example, the equilibrium vapor pressure of He at 1 K may be read off Fig. 10.4 as $p \cong 0.1 \text{ torr} \cong 133 \text{ dyn cm}^{-2}$. At this temperature, $\tau = 1.38 \times 10^{-16} \text{ erg}$ and, with the mass of a He atom about 4 amu, $n_Q = 1.88 \times 10^{20} \text{ cm}^{-3} \times 4^{3/2}$ (see Supplementary Material to Chapter 3). With these, $\mu_{\text{He}}(1 \text{ K}) \cong -1.0 \times 10^{-15} \text{ erg}$.

Chemical potentials are always expressed relative to some zero. In our example the chemical potential is relative to the state of zero kinetic energy of the He atoms.

When two phases of a substance are not in equilibrium, the stablest of the phases (lowest chemical potential) has the lowest equilibrium vapor pressure. Suppose that both phases are present in an enclosed volume, and that at least the less stable phase has a non-negligible vapor pressure. Then one mechanism by which the less stable phase can actually transform into the more stable phase is by evaporation, followed by condensation of the vapor onto the more stable phase.